

Madhusmita Talukdar

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Projects

Dowel | Power BI, Data Analytics & Modelling

March 2026

- Developed an AI-powered data intelligence platform that translates natural language queries into executable SQL, enabling non-technical users to extract insights from diverse data sources such as Excel, CSV, databases, APIs, and web data.
- Implemented an automated machine learning pipeline that performs schema understanding and selects appropriate models (Prophet, XGBoost for prediction, Isolation Forest for anomaly detection) to generate accurate, data-driven insights.
- Integrated probabilistic what-if analysis using Monte Carlo simulation and built interactive dashboards with real-time alerts, enabling faster decision-making and proactive anomaly detection.
- Repository Link: <https://github.com/Madhusmita111/Dowel>

Reddit–Google Trends Engagement Analysis | Power BI, Data Analytics & Modelling

December 2025

- Built an interactive Power BI dashboard integrating Google Trends and Reddit data to analyse relationships between search demand and online discussion engagement across multiple domains.
- Designed relational data models and implemented advanced DAX measures to calculate growth trends, engagement metrics, and time-based comparisons.
- Performed correlation analysis and developed KPI-driven visualizations to support trend forecasting and business-focused analytical insights.
- Repository Link: <https://github.com/Madhusmita111/powerbi-dashboard>

Crime Data Analysis & Prediction System | Python, Pandas, NumPy, Matplotlib, Seaborn, Machine Learning

May 2025

- Processed and analyzed 250K+ crime records using Pandas, performing data cleaning, feature engineering, and exploratory data analysis to prepare high-quality datasets for modeling.
- Applied machine learning techniques (clustering and pattern analysis) to identify crime hotspots, high-risk zones, and temporal trends, enabling predictive insights for urban safety planning.
- Built data visualizations and analytical dashboards to interpret model outputs and communicate actionable insights for data-driven decision-making.
- Repository Link: <https://github.com/Madhusmita111/Analyzing-Crime-Data-for-City-Safety>

Training

Smart URL Safety Checker | Python, Scikit-learn, XGBoost, Random Forest, SVM, Pandas, NumPy

May 2025

- Developed a machine learning system to classify malicious and legitimate URLs using models such as Logistic Regression, Decision Trees, Random Forest, SVM, and XGBoost.
- Engineered and selected key URL-based features (length, entropy, domain patterns) to improve model performance and robustness.
- Performed comparative evaluation using precision, recall, and F1-score to identify the best-performing model, with deployment-ready design for real-time web integration.
- Repository Link: <https://github.com/Madhusmita111/smart-url-safety-checker>

Certificates / Certifications

SQL (Intermediate) - Hackerrank | certificate link

October 2025

Human Computer Interaction – NPTEL | certificate link

May 2025

Skills

Language: Python, SQL, C/C++

Data Science & Analytics: Data Analysis, Statistical Modelling, Machine Learning, Data Visualization, Data Wrangling

Database: MySQL, PostgreSQL

Tools & Platforms: Excel, Power BI, Tableau, Jupyter Notebook, Google Colab, Git/GitHub, Linux, Jira, Docker

Other Skills: Agentic AI, Feature Engineering, Model Deployment, Data Pipeline Development, Agile Methodologies

Soft Skills: Analytical Thinking, Problem-Solving, Work ethic, Adaptability, Flexibility, Teamwork, Communication

Education

Lovely Professional University

2023 – Present

Bachelor of Technology in Computer Science and Engineering — CGPA: 7.83

Phagwara, Punjab

Spectrum Gurukul

2020 – 2021

Intermediate — Percentage: 92.60%

Nalbari, Assam

Shankardev Sishu/Vidya Niketan

2018 – 2019

Matriculation — Percentage: 92.80%

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